1. Business Background & Problem Statement

Turtle Games is a global game manufacturer and retailer selling books, board games, video games, and toys, both self-manufactured and sourced.

Situation: the company holds rich sales and customer review data but has not yet leveraged it.

Complication: currently has no data-driven understanding of customer behaviour, loyalty point accumulation, or segmentation to improve sales performance.

Question: how can customer and review data be analysed to improve sales performance?

Using the SCQA (Situation–Complication–Question–Answer) problem-solving framework, this report declares the problem and frames the subsequent Python and R analysis, covering regression, clustering, NLP, and statistical modelling as the **Answer**: a data-driven foundation for targeted marketing decisions.

2. Analytical Approach

Data Import and Cleaning (Python): The turtle_reviews.csv dataset (2,000 rows, 11 columns) was imported using pandas, then sense-checked with .info(), .isna().sum(), and .describe() to confirm no missing values and identify loyalty points' right-skew. Columns like language and platform were dropped as every row shared the same value and they carried no analytical importance, and columns were renamed for clarity (remuneration, spending_score). The cleaned data was exported as turtle_reviews_clean.csv, to be a reusable source for subsequent activity in both Python and R. This was done to ensure consistency across the full analysis.

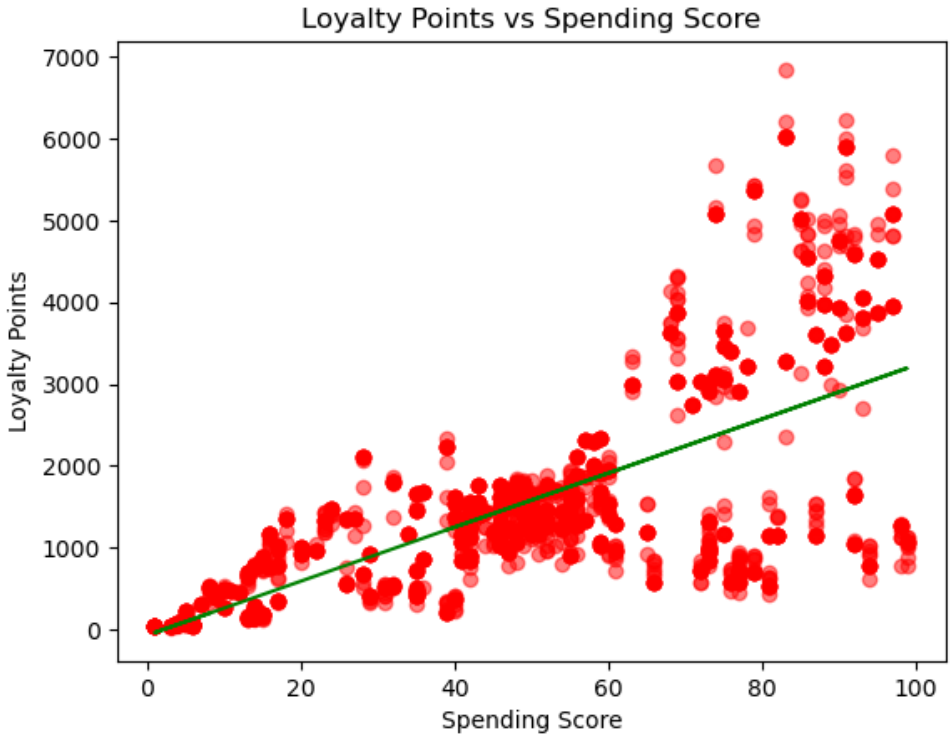
Python Analysis: scikit-learn's LinearRegression was used to test loyalty points against age, remuneration, and spending score individually to establish baseline relationships. A DecisionTreeRegressor then combined all three variables, using a 70:30 train-test split (random_state=42 for reproducibility) and pre-pruning (max_depth=4, min_samples_leaf=5) to prevent overfitting, chosen specifically because linear regression could not capture the non-linear, segment-dependent structure that the scatterplots revealed. KMeans clustering was applied using the Elbow and Silhouette methods together, rather than either alone, to justify the chosen number of clusters. For text analysis, NLTK and TextBlob were used for cleaning, tokenisation, and sentiment polarity scoring on the review and summary columns, with deduplication performed before analysis.

R Analysis: The same cleaned CSV was imported into R using tidyverse for data analysis and wrangling. dplyr's group_by()/summarise() quantified loyalty point differences across gender and education, while ggplot2 was used to produced exploratory visualisations. A multiple linear regression was applied to combine remuneration and spending score with age excluded based on its consistently negligible explanatory power across prior analysis. car::vif() confirmed no multicollinearity, and base R diagnostic plots validated model assumptions.

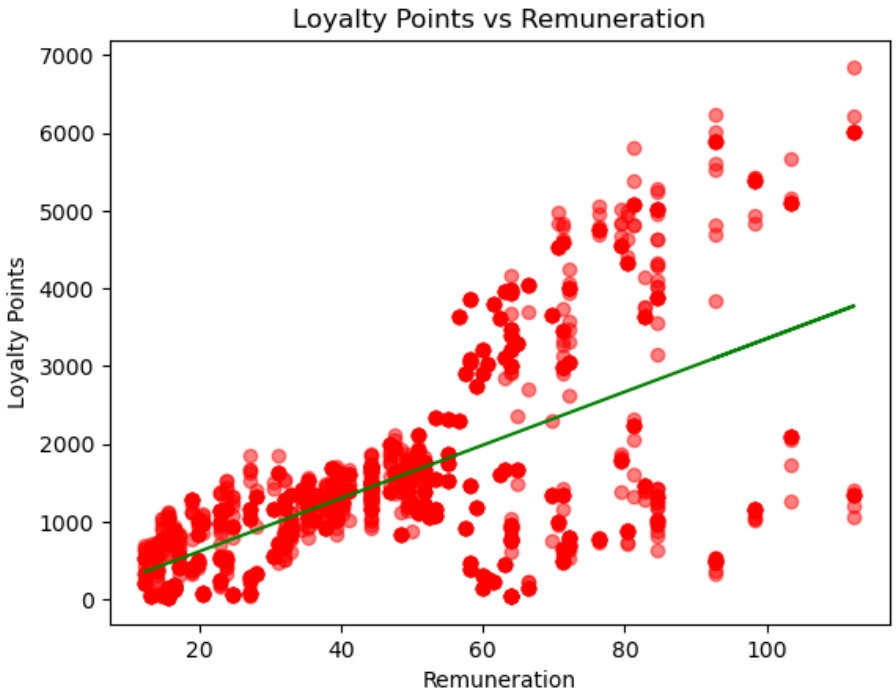
3. Visualisation and Insights

Visualisations were sequenced to build a narrative. The first was to establish if relationships exist (scatterplots), then why linear models fail to capture them (the two-band pattern), then how customers can be grouped (cluster plots), and finally what qualitative signal (text/sentiment) and *cross-validation* (R replication, model diagnostics) add to the picture.

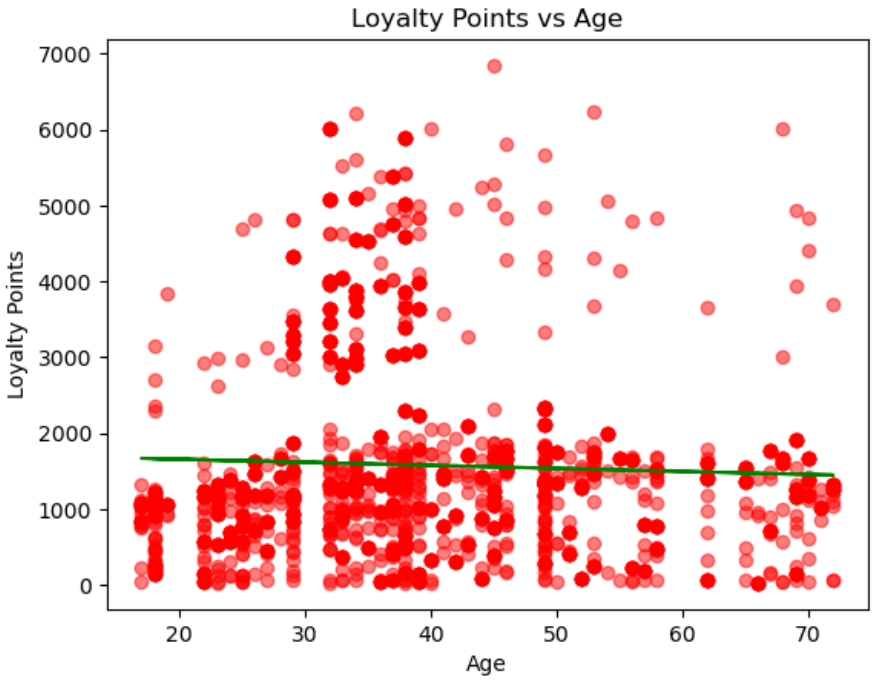
Scatterplots of loyalty points against spending score, remuneration, and age were chosen first because a scatterplot is the most direct way to visually assess linearity before committing to a model type. The spending score plot revealed a genuine two-band pattern: a lower, gently rising group and a separate, steeply rising high-spend group invisible from R^2 alone. An initial limitation was overplotting from 2,000 points; this was refined using alpha=0.5 transparency, letting point density reveal cluster concentration.



- The scatter plot does not show one clean, uniform linear cloud. Instead, there appears to be two distinct bands: a lower band of customers (spending score up to around 60) whose loyalty points rise gently and stay under 2,000, and a separate, steeper band of customers (spending score above 60+) whose loyalty points jump sharply higher, several reaching 5,000–7,000.
- This suggests the population may contain at least two different customer segments rather than a single group, and a simple linear model averages across both, which is likely why the fit is only moderate rather than strong.

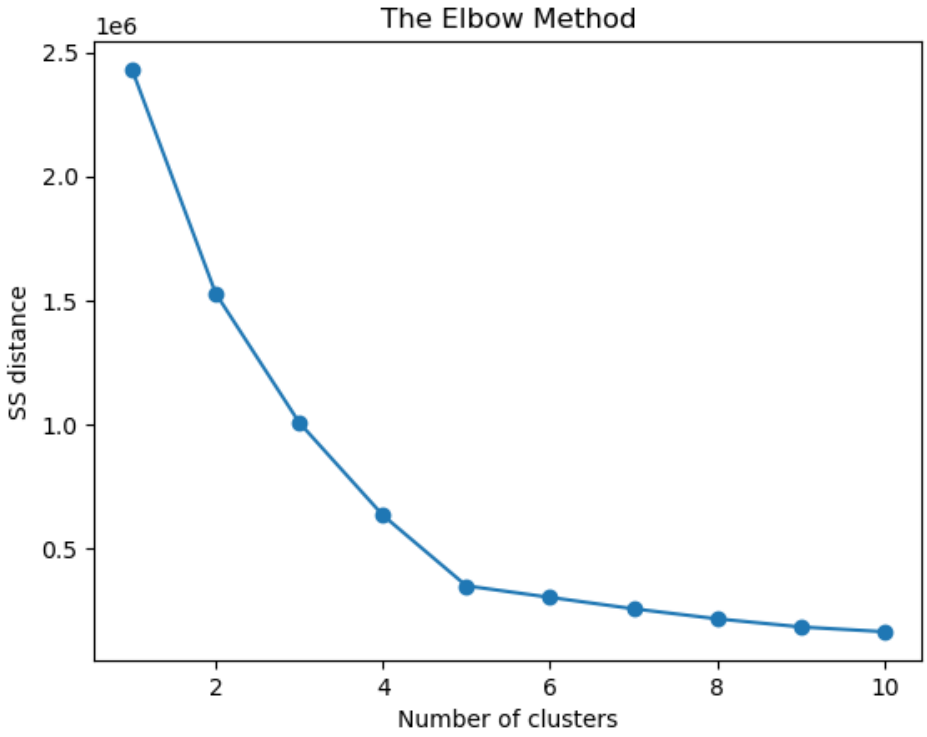


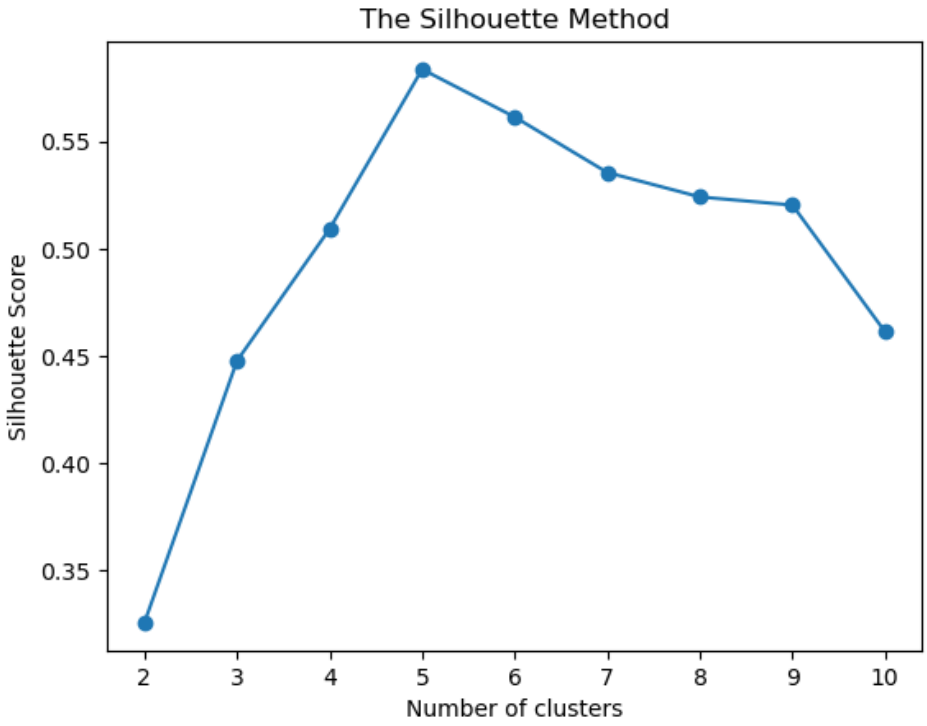
- The scatter shows a positive trend at lower-to-mid remuneration levels, but the spread widens considerably as remuneration increases. At remuneration above 80k£, points range anywhere from under 1,000 to over 6,000 for similar remuneration values. This unevenness means the linear model is a much weaker predictor for higher earners specifically, even though it captures some signal at the lower end.



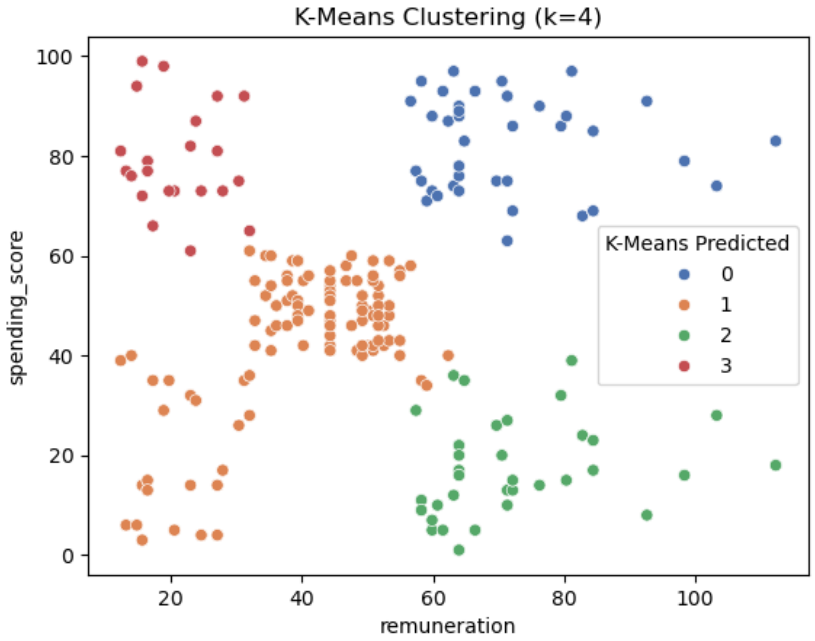
- The scatter plot confirms the previous observations for the coefficient visually: points are scattered broadly across all loyalty point values at every age, with no visible upward or downward drift. The regression line is essentially flat. Age alone is not a useful predictor of loyalty points.

The Elbow and Silhouette plots were paired, since the Elbow method alone requires subjective judgement about where the "bend" occurs, while Silhouette gives a clear peak, both independently confirming k=5.

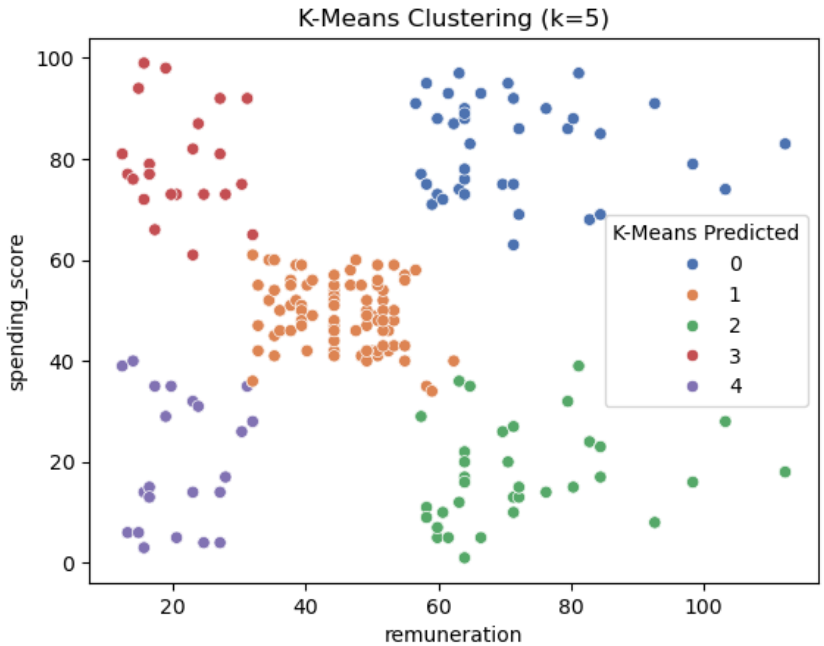




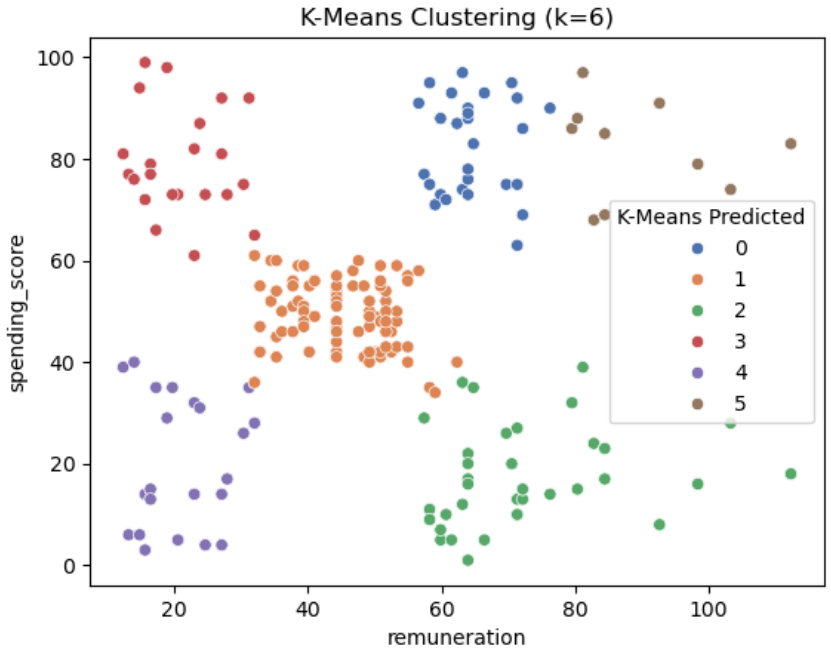
The k=4/5/6 comparison scatterplots visually justified the final choice by showing k=5 as the cleanest separation, with k=4 merging distinct groups and k=6 arbitrarily splitting one.



Four groups are visible: a low-remuneration/high-spend group (red, top-left), a low-remuneration/low-spend group merged together with the central-middle group (orange, occupying a large central-to-left blob), a high-remuneration/low-spend group (green, bottom-right), and a high-remuneration/high-spend group (blue, top-right). The orange group is doing double duty here as it's covering two visually distinct areas (the low-spend cluster on the left and the dense central cluster) that don't obviously belong together. That's a sign this k is too coarse as genuinely different customer types are being forced into one label.



This plot is visibly much cleaner. There are five distinct blobs with gaps between them: high remuneration + high spend (blue, top-right), high remuneration + low spend (green, bottom-right), low remuneration + high spend (red, top-left), low remuneration + low spend (purple, bottom-left), and a distinct middle-income/middle-spend cluster sitting in the centre (orange). Every group now occupies its own clearly bounded region with visible white space between clusters.



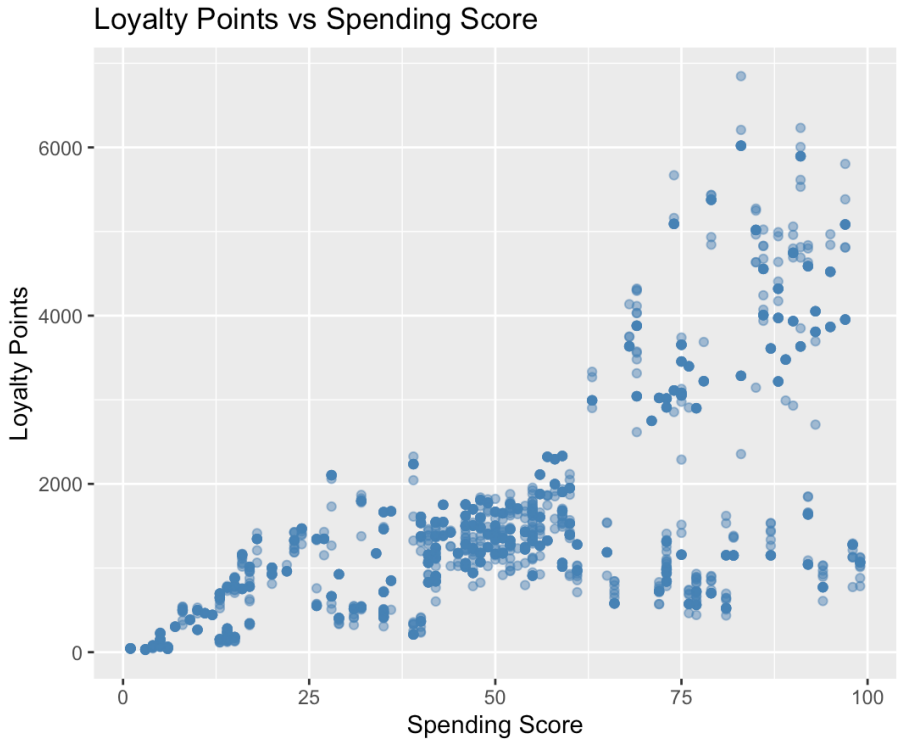
The five groups from k=5 are still here, but the high-remuneration/high-spend group (previously all blue) has been split into two colours (blue and brown) with no visible gap between them, as they occupy the same continuous region of the plot. This is an over-fragmentation flag: k=6 isn't finding a genuinely new customer type, it's just arbitrarily cutting one real group in half.

Word clouds were refined because the initial raw version was dominated by meaningless filler words; the cleaned version showed informative terms such as "game", "fun", "great". Their key limitation, however, is that the size of words is tied to frequency and they convey no sentiment, which is why polarity histograms were added alongside them.

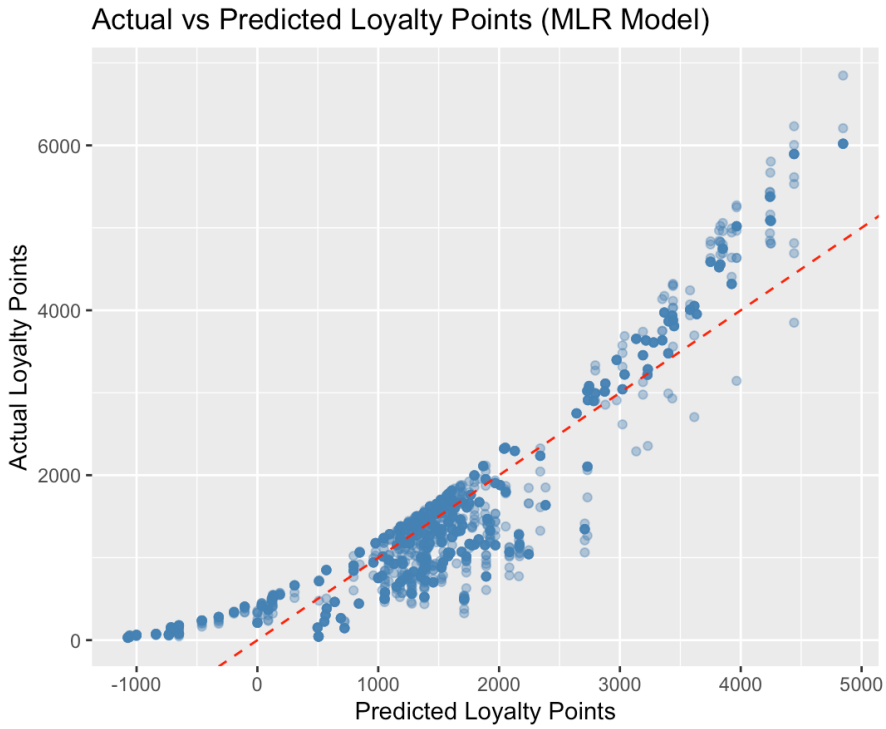


Visualisation showing Word cloud after stopword removal

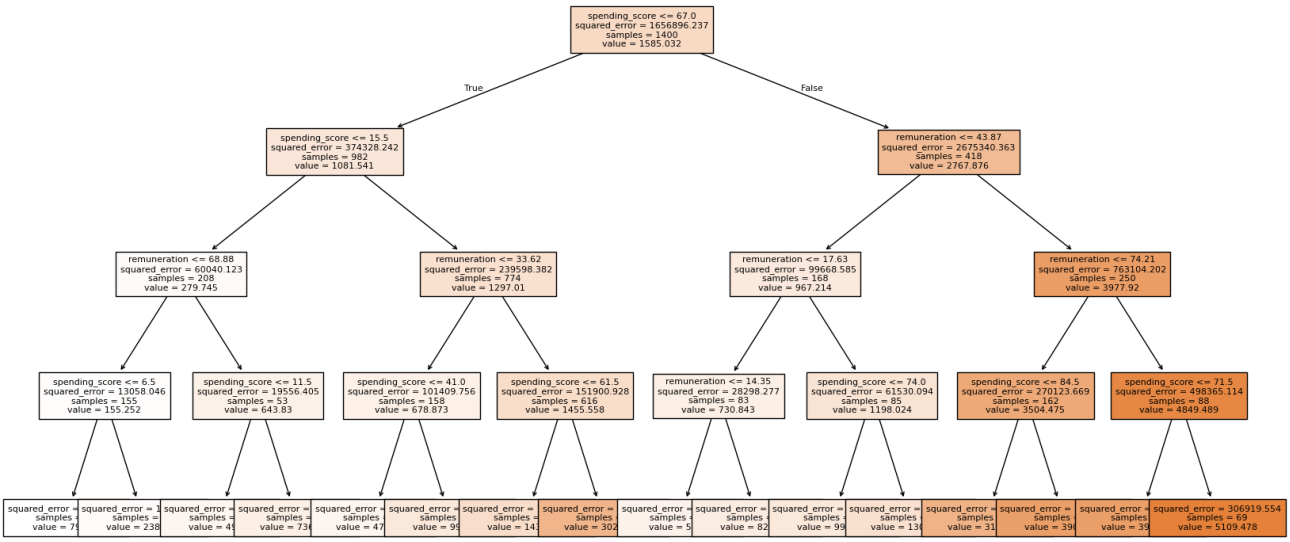
R's ggplot2 scatterplot for loyalty vs spending score independently reproduced the two-band pattern shown earlier in Python visualisation, further strengthening confidence.



Although the MLR model achieved a strong R^2, the actual-versus-predicted plot and residual diagnostics showed reduced predictive accuracy at higher values, a weakness that was evident only through visualisation and not from the summary statistic alone.



The pruned decision tree was included specifically because it is the only visualisation in this analysis that shows a model's actual decision-making logic, not just its output. Unlike a scatterplot or regression line, it lets us trace exactly how a prediction is formed. The root split on spending score (≤67 vs >67) confirms it as the dominant driver, with remuneration acting as a consistent secondary factor at every subsequent branch. Its main limitation as a visual is readability at depth: an unpruned tree's plot was too dense to interpret, which is part of the justification for pruning.



Before pruning, there was a noticeable gap between training and test R^2, which can be said to be the classic overfitting. After pruning, that gap essentially disappears: training $R^2 = 0.946$, test $R^2 = 0.943$. This tells us that the pruned tree isn't memorising the training data; it has found a genuine, reusable pattern that holds up on customers it's never seen before.

4. Patterns, Predictions & Recommendations

Analysis revealed three consistent patterns. First, loyalty points increase in a non-linear and segment-dependent manner**.** Customers don't accumulate points at a constant rate as spending or remuneration increases. The decision tree model ($R^2 = 0.943$) identified clear spending and remuneration thresholds that separate customers into distinct groups, indicating that high-spending customers earn loyalty points at a faster rate.

Second, loyalty points are right-skewed**,** with the mean (1,578) exceeding the median (1,276). This results from a small number of customers getting very high loyalty scores. Also, linear regression models tend to underestimate high-value customers and can produce unrealistic predictions at the extremes of the data. The actual-versus-predicted plot confirmed that prediction accuracy declines for these customers.

Third, age has no real influence on loyalty points, demonstrated by a near-zero correlation (-0.04), a negligible R^2 (0.0018), and the absence of age-based decision tree splits. Review sentiment showed positive and negative reviews, with recurring

complaints relating to incomplete game kits and product quality indicating that customer satisfaction is influenced by product experience.

Based on these findings, the pruned decision tree should be adopted as the primary forecasting model because it predicts loyalty points more accurately than multiple linear regression. The business should also target the high-remuneration, low-spending customer segment with personalised promotions, as this group represents a significant opportunity to increase spending and revenue.

Appendix

Appendix A — Analytical Approach: Supporting Detail

A.1 Data requirements breakdown, by business question

Business Question	Required Features	Source Columns	Cleaning Required
Loyalty point accumulation	Numeric, behavioural variables	age, remuneration, spending_score, loyalty_points	Column rename only; no missing values present
Customer segmentation	Two continuous, comparably scaled variables	remuneration, spending_score	Dropped to two-column subset; remuneration ranging from ~£10k–115k and spending_score between 1–100.
Text data for marketing	Unstructured free text	review, summary	Lowercasing, punctuation stripping, deduplication, tokenisation, stopword removal

A.2 Rationale and limitations, technique by technique

Simple Linear Regression: This was done first because it is the simplest possible model. The goal was to establish a baseline before justifying any added complexity. Its core limitation (confirmed by R^2 values of only 0.452, 0.380, and 0.0018 for spending, remuneration, and age respectively) is the assumption that each variable has one fixed, constant effect on loyalty points regardless of other conditions. The two-band pattern visible in the spending-score scatterplot is direct visual proof this assumption is false for this dataset: the effect of spending on loyalty points differs depending on which underlying customer segment a person belongs to.

Decision Tree Regressor: This was applied to overcome the linear model's inability to represent conditional, segment-dependent effects. A tree naturally partitions customers

into subgroups, allowing different effective rules for different segments. The max_depth=4, min_samples_leaf=5, min_samples_split=5 pruning parameters were selected by first fitting an unpruned tree, and then tightening constraints until training and test R^2 converged (0.946 vs 0.943).

KMeans Clustering: The Elbow method alone is often ambiguous, as questions like "where exactly did the bend occur" come up. Running the Silhouette method in parallel and finding both independently agreed on k=5 strengthens confidence in the result beyond what either method would justify alone.

NLTK/TextBlob sentiment analysis: A pre-built lexicon approach was chosen because it requires no labelled training data (Turtle Games' reviews have no existing "positive/negative" labels to learn from). Limitation, demonstrated directly in the findings: lexicon-based scoring evaluates words with limited surrounding context, so a neutral product-category noun like "game" can receive an unexpectedly negative polarity score purely due to how it appears in the lexicon's training corpus, independent of actual customer sentiment.

Descriptive statistics and distribution analysis: mean, median, skewness, and visual histogram inspection were used together rather than any single statistic alone, since skewness alone doesn't reveal *where* the skew concentrates. The secondary "hump" around 3,000–4,000 loyalty points visible in the histogram would be invisible from summary statistics alone, yet is directly relevant to the business, since it hints at the same high-value segment identified independently through clustering.

Appendix B: Recommendations, Business Question Responses, and Process Reflection

B.1 Direct responses to Turtle Games' four original questions

Business Question	Finding	Recommendation
How do customers engage with and accumulate loyalty points?	Spending score is the dominant driver (root split in decision tree), followed by remuneration; age has no effect. The relationship is non-linear as customers above a spending threshold accumulate points at a disproportionately higher rate than the overall average suggests.	Redesign a loyalty programme around spending-based tiers rather than a flat accrual rate, since the data shows customer behaviour already naturally clusters into distinct tiers.
How can customers be segmented, and	Five statistically justified segments exist (k=5, confirmed by both Elbow and Silhouette methods): one large "average"	Prioritise the high-remuneration/low-spend segment (330 customers) for targeted upsell campaigns as

which groups targeted?	group (774 customers) and four smaller, more distinct groups defined by high/low combinations of remuneration and spending.	this group has the financial capacity but is not currently converting it into spend, representing the clearest near-term revenue opportunity.
How can text data inform marketing campaigns?	Sentiment is mildly positive; specific, recurring, fixable complaints were identified (incomplete kits, build quality, age-inappropriate instructions for certain products); positive reviews cluster strongly around gift-giving and family/educational use.	Feed recurring negative-review themes directly to product quality control; lean into gift-giving and family-learning messaging in marketing campaigns, since this is where the most enthusiastic organic sentiment already concentrates.
Is the loyalty points data suitable for predictive modelling (distribution, skew)?	The data is right-skewed (mean > median; confirmed in both Python and R), which contravenes the normality assumption underlying simple linear models demonstrated concretely by the MLR model producing an impossible negative prediction for a low-end customer.	Predictive modelling efforts should prioritise tree-based methods (which make no normality assumption) over linear regression; if linear methods are required for interpretability, apply a log-transformation to loyalty points first to reduce skew.